

MLP + Parallel Mini-Batch SGD (MNIST)

SINGLE THREADED

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1. Introduction

1.1. Problem

The MNIST dataset is a subset of a larger set available from NIST. It is a somewhat classic example and often considered to be the “Hello, World!” of machine learning. It’s a ‘classification’ problem in the sense that we are expected to train a neural network to be able to recognize handwritten digits.

1.2. Dataset Description

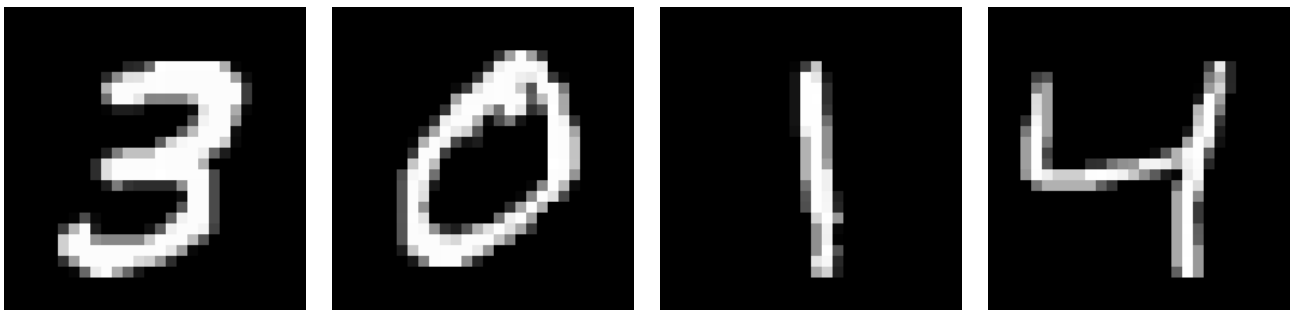


Figure 1: MNIST Dataset Samples

These are images with a very low resolution of 28×28 *greyscale* pixel values encoded on 8 bits (i.e. with the values of each pixel ranging from 0 to 255).

The dataset is split as follows:

```
dataset/MNIST/
├── t10k-images.idx3-ubyte  -> . Test images
├── t10k-labels.idx1-ubyte  -> . Test labels
├── train-images.idx3-ubyte  -> . Training images
└── train-labels.idx1-ubyte -> . Training labels
```

The 2 types of files:

1.2.1. Image files (idx3):

```
[0..3]  magic number = 0x00000803 (from byte 0 to byte 3 inclusive)
[4..7]  num_images
[8..11] num_rows   (28)
[12..15] num_cols  (28)
[16..]  pixels: uint8, row-major, one byte per pixel
```

1.2.2. Label files (idx1):

```
[0..3]  magic number = 0x00000801  
[4..7]  num_labels  
[8..]   labels: uint8, value 0-9
```

The magic number in each filetype serves as a sanity check

1.3. Motivation

2. Serial Implementation

2.1. Preprocessing

2.2. Algorithm Description & Complexity Analysis

3. Parallel Implementation

3.1. Preprocessing

3.2. Design Rational

3.3. Data/Task Decomposition

3.4. Synchronisation

4. Experimental Results

4.1. Timing Table

4.2. Speedp Graph

4.3. Accuracy Comparison

5. Conclusion and Perspectives

5.1. Findings

5.2. Limitations

5.3. Future Direction

6. References